MERT KOŞAN

PH.D. CANDIDATE

+90 533 358 25 02 / +1 805 568 88 95 | mertkosan@gmail.com | mertkosan.com | mertkosan | mertkosan

WORK EXPERIENCE

Ph.D. Machine Learning Scientist - Visa Research

Internship - Improving AI Decisions with Feedback Loop Active Learner

June 2022 – September 2022 | Austin, USA

- Proposed a human-in-the-loop recommender system improving AI decisions using active learning for anomaly detection applications, but applicable to different application scenarios such as credit card approval system. The framework helps increasing in precision and recall.
- Patent filing under progress. Writing a paper.

Ph.D. Machine Learning Scientist - Visa Research

Internship - Peer Group Analysis with Anomaly Detection

June 2021 – September 2021 | Austin, USA

- Proposed a scalable and parallelizable dynamic peer grouping algorithm working with any existing anomaly detection algorithm to reduce false alarms and detect anomalies in their early stages. The framework reduces the false positives substantially and detects multiple events earlier.
- Filed a patent, which is currently under review.

Ph.D. Machine Learning Scientist - Visa Research

Internship - Fraud Detection and Profiling

June 2020 – August 2020 | Austin, USA

- Developed a framework that generates profiling and optimized strategies for near-real time unsupervised fraud detection on graphs. The framework profiles the fraud automatically and accurately in less than a second with a small number of transactions.
- Filed a patent, which is currently under review.

Part-time Software Engineer - Proente

June 2017 – August 2018 | Istanbul, Turkey

Intern Software Engineer - Robert Bosch GmbH

June 2016 – August 2016 | Plochingen, Germany

TEACHING EXPERIENCE

Co-designer/Instructor - UC Santa Barbara

Machine Learning Workshops | Python

May 2019 – June 2019 | Santa Barbara, USA

- Helped the students, from A Self-Organized PhD Students Exchange Program by Junior Nanotech Network, for their projects.

Teaching Assistant - UC Santa Barbara

Data Structures & Algorithms | C++, Python

September 2018 – April 2019 | Santa Barbara, USA

Co-designer/Instructor - Sabanci University

Enhancement Workshops for Computer Science | Git, Linux, Python etc.

January 2018 – June 2018 | Istanbul, Turkey

Teaching Assistant - Sabanci University

Database Systems & Operating Systems | MySQL, C++, Java

February 2016 – May 2017 | Istanbul, Turkey

EDUCATION

Ph.D. in Computer Science

University of California at Santa Barbara

2018 – 2023 (Expected) | Santa Barbara, USA

Advisor: Prof. Ambuj K. Singh | GPA: 4.00

Research Area: **Network Science, Data Mining, Machine Learning, Anomaly Detection, Graph Neural Networks**

B.Sc. in Computer Science

Sabanci University

2013 – 2018 | Istanbul, Turkey

GPA: 3.99 | Ranked 1^{st}

RESEARCH PROJECTS

Robust Ante-hoc Graph Explainer

First author, Under Review, 2021-2022

- A novel robust ante-hoc graph explainer that utilizes graph neural networks and bilevel optimization. It enables the user to reproduce model behavior based on the explanations.
- Outperformed state-of-the-art graph classification, post-hoc, and ante-hoc explainer baselines in accuracy, meaningful explanations, reproducibility, and robustness.

Link Prediction without Graph Neural Networks

Second author, Under review, 2021-2022

- A novel topology-centric end-to-end architecture, consisting of graph learning, topological heuristics, and training with N-pair loss for the link prediction task.
- Outperformed state-of-the-art link prediction baselines with 145% more accuracy and 6000 times faster inference time.

Global Counterfactual Explainer

Mert Kosan; Zexi Huang* et al., "Global Counterfactual Explainer for Graph Neural Networks." ACM International Conference on Web Search and Data Mining (WSDM) 2023 (*:equal contribution) arXiv:2210.11695*

First co-author, WSDM'23, 2021-2022

- A novel problem of global counterfactual explanations for graph neural networks applied on graph classification tasks. Our method generates global counterfactual summary by vertex-reinforced random walks on an edit map.
- Outperformed state-of-the-art local counterfactual explainers in explanations, coverage, and recourse quality.

Event Detection on Dynamic Graphs

Mert Kosan et al., "Event detection on dynamic graphs." arXiv preprint arXiv:2110.12148 (2021).

First author, arXiv'21, 2019-2021

- An event detection framework using dynamic graph neural networks further enhanced by structural and temporal self-attention, tested on multiple new datasets.
- DyGED learns correlations between the graph macro dynamics (a sequence of graph-level representations) and labeled events.

SKILLS

- **Programming:** Python, C++, Java, Javascript, SQL
- **Machine Learning Libraries:** PyTorch, Keras, Scikit-Learn
- **Tools:** Neo4j, Git, LaTeX, Office, MySQL
- **Languages:** Turkish (native), English (fluent), Spanish (beginner)
- **Personality:** Result-oriented | Responsible | Hardworking
- **Hobbies:** Football | Strategy Games | Chess | Snooker

OTHER SELECTED PROJECTS

Bike Sharing

September 2018 – December 2018

- Designed and implemented a bike sharing rebalancing problem using a network flow approach.

Impact Measurement of Tweets

October 2017 – June 2018

- Measured the hidden impact of a tweet by detecting copy-pasted or referenced tweets named as “Hidden Retweets” by Dr. Arin.

Differential Privacy Framework for Movie Recommendation System

September 2016 – September 2017

- Proposed and implemented a movie recommendation system that utilize differential privacy framework which guarantees the privacy of the individuals without losing the utility.

CryCur

Command line tool for crypto-currency applications

January 2017 – May 2017

Economics and Terrorism

Correlation of economic prospects (e.g. GDP) and terror incidents

January 2017 – May 2017

MORE RESEARCH PROJECTS

Multi-scale Graph Anomaly Detection

Second author, 2021-Present

- Detecting multi-scale unsupervised graph anomalies based on connectivity patterns and node attributes using recurrent graph neural networks with reconstruction error.

Feature-based Individual Fairness in k-Clustering

Kar, Debajyoti, et al. "Feature-based Individual Fairness in k-Clustering." arXiv preprint arXiv:2109.04554 (2021).

Second Author, 2022-Present

- Proposing a new fairness metric that measures the fairness score of each individual in clusters decided by k-means.
- The individual should have a similar person (not necessarily in terms of protected attributes) like themselves in the group so that they can share information better.

Cluster Meta-Learning for Network Edge Estimation

Third Author, 2022-Present

- Predicting missing values in flow networks using meta-learning (where each cluster is sub-task) for node regression task with line graphs.

Black Box Modeling in Decision Making

First Author, 2019-Present

- Analyzing how the risky behavior of individuals alters in the group setting from the individual setting using a black-box model comparing white-box models (e.g. Prospect Theory).

ACADEMIC SERVICES

- **Registration Chair:** KDD 23’
- **Academic Paper Reviwer:** NeurIPS, ICLR, KDD, WebConf, WSDM, ICDM, SDM, AAAI, TIST, TKDD, TKDE.